Chapter 1

Probability Models

Probability models offer a mathematical framework for describing and understanding random phenomena.

Examples of random phenomena include:

- the outcome of a die roll
- the lifetime of a computer chip
- the time until the next bus arrives
- the price of a stock one year from now

Probability models will be used in advanced courses that you will study at Heriot-Watt, including Stochastic Processes (F79SP), Survival Models (F79SU), Risk Theory (F70RT) etc.

Key point: A solid grasp of probability theory is essential for success in many advanced courses.

1.1 Formal definition of a probability model

We begin with the formal, mathematical definition of a probability model. This material is quite abstract, but we want to understand the abstract foundations before moving on to practical applications.

A probability model has 3 components, the sample space Ω, the event space $\mathcal{F}$ and the probability measure $\mathbb{P}$.

1. Sample space Ω (or sometimes S)
   - Ω is the set of all possible outcomes of a random experiment.
   - We use the symbol ω to denote an element of Ω, and we call ω a sample point.

   Example 1.1. Roll a die, $\Omega = \{1, 2, 3, 4, 5, 6\}$. One possible outcome is $\omega = 4$, meaning that we roll a 4.

   Example 1.2. Measure lifetime of a computer chip. Since (theoretically) the lifetime could be any non-negative number, we take $\Omega = [0, \infty)$.

   Example 1.3. Two tosses of a coin, $\Omega = \{\text{HH,HT,TH,TT}\}$, where H and T represent the Head and Tail of a coin.

2. Event space $\mathcal{F}$

   - $\mathcal{F}$ is a collection of subsets of the sample space Ω, which satisfies the following properties:

(a) $\Omega \in \mathcal{F}$;
(b) if $A \in \mathcal{F}$ (where A is a subset of Ω), then $\Omega \setminus A = A^c \in \mathcal{F}$;
(c) if $A_1, A_2, \cdots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$ (countable unions are in $\mathcal{F}$).

- Each subset in the collection $\mathcal{F}$ is called an event.

A collection of subsets satisfying properties (a)–(c) above is called a **σ-algebra** or **σ-field**.

Example 1.1 continued. Roll a die, $\Omega = \{1, 2, 3, 4, 5, 6\}$. Then 'roll an even number' is represented as $A = \{2, 4, 6\} \subset \Omega$, and we can define our event space $\mathcal{F}$ to include A as one of its elements, so that 'roll an even number' is an event; that is, $A \in \mathcal{F}$. By property (b), the event space $\mathcal{F}$ must then also include the set $A^c = \{1, 3, 5\}$, which is the event 'roll an odd number'.

In the definition of an event space we require that the whole state space should be an event and that we can combine events by asking that A or B is an event if $A, B \in \mathcal{F}$. There are additional ways to combine events. For instance, the empty set is considered to be an event, i.e. nothing happening is an event. Also if $A, B \in \mathcal{F}$ are events then the event that both A and B occur is also an event. Both of these follow from the definition of the event space.

Lemma 1.1. *Let $\mathcal{F}$ be an event space then*

- $\emptyset \in \mathcal{F}$
- *If $A_1, A_2, \cdots \in \mathcal{F}$, then $\bigcap_i A_i \in \mathcal{F}$*

Proof

- By property (a), $\Omega \in \mathcal{F}$. Then by property (b), $\Omega^c \in \mathcal{F}$; that is, $\emptyset \in \mathcal{F}$, as required.
- If $A_1, A_2, \cdots \in \mathcal{F}$, then

$$\begin{aligned}
&\text{By Property (b)} \Rightarrow A_1^c, A_2^c, \cdots \in \mathcal{F} \\
&\text{By Property (c)} \Rightarrow \bigcup_i A_i^c \in \mathcal{F} \\
&\text{By Property (b)} \Rightarrow \left(\bigcap_i A_i\right)^c = \bigcup_i A_i^c \quad \text{(de Morgan's Laws from set theory)}
\end{aligned}$$

□

Example 1.1 continued. In the example of rolling a die, 'Roll a multiple of 3' is represented as $B = \{3, 6\}$, and we can define $\mathcal{F}$ to include B; that is, $B \in \mathcal{F}$ and 'roll a multiple of 3' is an event. Then by property (c), $A \cup B = \{2, 3, 4, 6\} \in \mathcal{F}$ is also an event. (Notice that property (c) as originally stated applies to an infinite sequence of events, but we just need to take $A_3 = A_4 = \cdots = \emptyset$ to be able to apply property (c) to the union of just 2 events.) Similarly, $A \cap B = \{6\}$ is an event.

It's natural in the above example to take $\mathcal{F}$ to consist of all subsets of Ω, and then 'an event' simply means any subset of Ω. We will see later that in more complicated examples, this is not possible.

3. Probability measure $\mathbb{P}$

   A probability measure $\mathbb{P}$ is a *function* $\mathbb{P} : \mathcal{F} \to \mathbb{R}$ defined on the event space $\mathcal{F}$ which satisfies the following properties:

   (I) $\forall A \in \mathcal{F},\ \mathbb{P}(A) \geqslant 0$

   (II) $\mathbb{P}(\Omega) = 1$

   (III) For any *mutually exclusive* events $A_1, A_2, \cdots \in \mathcal{F}$ we have $\mathbb{P}(\bigcup A_i) = \sum \mathbb{P}(A_i)$.
   Note: The events $A_1, A_2, \cdots \in \mathcal{F}$ are *mutually exclusive* if for any two of the events A_i and A_j with $i \neq j$, we have $A_i \cap A_j = \emptyset$.

   The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a probability model (or probability space).

   Interpretation of $\mathbb{P}(A)$

   The quantity $\mathbb{P}(A)$, the probability of event A, represents how likely it is that the outcome of the random experiment lies in A. This likelihood is measured on a scale from 0 to 1.

   Example 1.1 continued. Roll a die, $\Omega = \{1, 2, 3, 4, 5, 6\}$. Take the event space to be $\mathcal{F}$ = all subsets of Ω. If the die is fair, then $\mathbb{P}(\{1\}) = \mathbb{P}(\{2\}) = \mathbb{P}(\{3\}) = \mathbb{P}(\{4\}) = \mathbb{P}(\{5\}) = \mathbb{P}(\{6\})$. From property (III) and property (II) it then follows that $\mathbb{P}(\{\omega\}) = \frac{1}{6}$ $\forall \omega \in \Omega$. Applying property (III), then

$$\begin{aligned} \mathbb{P}(\text{roll an even number}) &= \mathbb{P}(\{2, 4, 6\}) \\ &= \mathbb{P}(\{2\}) + \mathbb{P}(\{4\}) + \mathbb{P}(\{6\}) \\ &= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2} \end{aligned}$$

1.2 Properties of a probability measure $\mathbb{P}$

Using the defining properties of a probability measure $\mathbb{P}$, we can deduce other important properties that are useful for computing probability values.

1. Law of Complements.

   If $A \in \mathcal{F}$, then $\mathbb{P}(A) = 1 - \mathbb{P}(A^c)$

2. Inclusion–Exclusion.

   If $A, B \in \mathcal{F}$, then $\mathbb{P}(A \bigcup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \bigcap B)$

3. Monotone Property of Probability Measures

   If $A, B \in \mathcal{F}$ and $A \subseteq B$, then $\mathbb{P}(A) \leqslant \mathbb{P}(B)$

Proofs

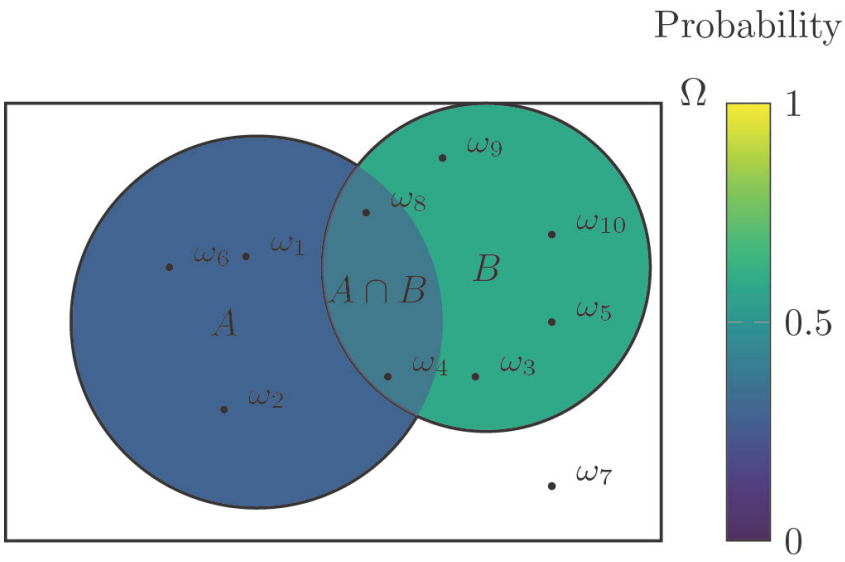


Figure 1.1: Illustration of a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ showing events A, B and their intersection $A \cap B$. Individual outcomes ω_i are represented as points within the sample space Ω. The colour bar encodes probability values of the events A, B and $A \cap B$, with shading corresponding to event likelihoods.

1. For any $A \in \mathcal{F}$, we have $A^c \cap A = \emptyset$, so
$$\begin{aligned} \mathbb{P}(A^c \cup A) &= \mathbb{P}(A^c) + \mathbb{P}(A) && \text{by property (III)} \\ \Rightarrow \quad \mathbb{P}(\Omega) &= \mathbb{P}(A^c) + \mathbb{P}(A) \\ \Rightarrow \quad 1 &= \mathbb{P}(A^c) + \mathbb{P}(A) && \text{by property (II)} \\ \Rightarrow \quad \mathbb{P}(A^c) &= 1 - \mathbb{P}(A) \end{aligned}$$
2. For any $A, B \in \mathcal{F}$, notice that
$$A \cup B = A \cup (B \setminus A)$$
But $A \cap (B \setminus A) = \emptyset$, a disjoint union, so we can now apply property (III):
$$\begin{aligned} \mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B \setminus A) \\ \Rightarrow \quad \mathbb{P}(B \setminus A) &= \mathbb{P}(A \cup B) - \mathbb{P}(A) \end{aligned}$$
Next, notice that
$$\begin{aligned} B &= (A \cap B) \cup (B \setminus A) && \text{(disjoint union)} \\ \Rightarrow \quad \mathbb{P}(B) &= \mathbb{P}(A \cap B) + \mathbb{P}(B \setminus A) && \text{property (III)} \\ &= \mathbb{P}(A \cap B) + (\mathbb{P}(A \cup B) - \mathbb{P}(A)) \\ \Rightarrow \quad \mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \end{aligned}$$
3. For any $A, B \in \mathcal{F}$ with $A \subseteq B$,
$$\begin{aligned} B &= A \cup (B \setminus A) && \text{(disjoint union)} \\ \Rightarrow \quad \mathbb{P}(B) &= \mathbb{P}(A) + \mathbb{P}(B \setminus A) && \text{property (III)} \\ \Rightarrow \quad \mathbb{P}(B) &\geqslant \mathbb{P}(A) && \text{property (I) applied to } B \setminus A \end{aligned}$$

1.3 Some examples and discussion

Example 1.1 continued. Random experiment: Roll a die.

Probability model:

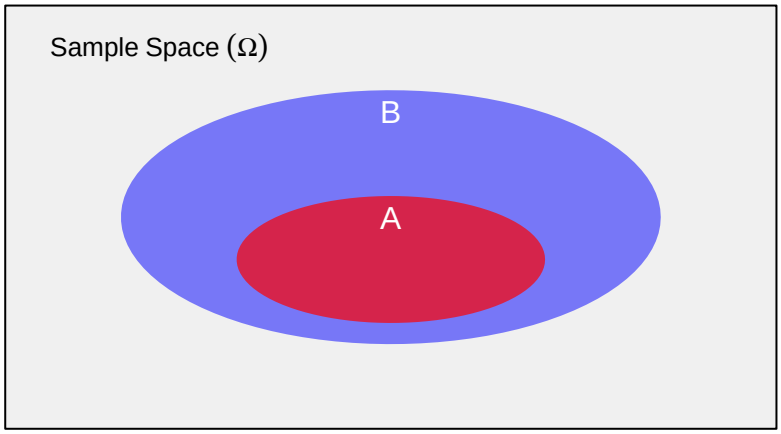


Figure 1.2: Illustration of the monotonicity property of probability measures. The red region represents event A, and the blue region represents event B with $A \subseteq B$. By the monotonicity property, we have $\mathbb{P}(A) \leq \mathbb{P}(B)$.

- Sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$
- Event space $\mathcal{F}$ = all subsets of Ω
- Specify probability measure $\mathbb{P}$ by specifying the value of $\mathbb{P}(\{\omega\})$ for every $\omega \in \Omega$

Since the sample space can be written as the disjoint union $\Omega = \{1\} \cup \{2\} \cup \{3\} \cup \{4\} \cup \{5\} \cup \{6\}$, then

$$\left.\begin{array}{l}\text{Property (III)} \Rightarrow \mathbb{P}(\Omega) = \mathbb{P}(1) + \mathbb{P}(2) + \cdots + \mathbb{P}(6) \\ \text{Property (II)} \ \Rightarrow \mathbb{P}(\Omega) = 1\end{array}\right\} \Rightarrow \begin{array}{l}\text{the probability measure must satisfy} \\ \mathbb{P}(1) + \mathbb{P}(2) + \cdots + \mathbb{P}(6) = 1\end{array}$$

For example, in a biased die, we might assign probabilities as follows:

$$\mathbb{P}(1) = \mathbb{P}(2) = \mathbb{P}(3) = \mathbb{P}(4) = \mathbb{P}(5) = \frac{1}{10} \quad \text{and} \quad \mathbb{P}(6) = \frac{1}{2} \qquad (\textit{biased} \text{ die})$$

Once we have specified the values of $\mathbb{P}(\omega)$ $\forall \omega \in \Omega$, we can use Property (III) to compute $\mathbb{P}(A)$ for other $A \in \mathcal{F}$.

Suppose $A = \{2, 4, 6\} = \{\text{outcome on the roll is even}\}$, then

$$\mathbb{P}(A) = \mathbb{P}(2) + \mathbb{P}(4) + \mathbb{P}(6) = \frac{7}{10} \text{ by property (III)}$$

If the sample space Ω is finite, then we can take $\mathcal{F}$ to be the power set of Ω i.e. $\mathcal{F}$ = (all subsets of Ω) and we can define the probability measure $\mathbb{P}$ by specifying $\mathbb{P}(\{\omega\})$ $\forall \omega \in \Omega$ s.t. $\sum_{\omega \in \Omega} \mathbb{P}(\{\omega\}) = 1$.

Remark 1.1. Recall a set Ω is countable if there exists an injective map $\psi : \Omega \to \mathbb{N}$. Examples of countable sets are: any finite set, the natural numbers $\mathbb{N}$, integers $\mathbb{Z}$ and the rational numbers $\mathbb{Q}$. Examples of sets that are not countable are non-trivial intervals $[a, b]$ with $a < b$ and the set of real numbers $\mathbb{R}$.

Note: You are expected to have learned the fundamental *counting principle* such as **combinations** and **permutations** before taking this course.

Example 1.4. Random experiment: Flip a coin until you obtain a 'Head'.

Probability model:

- Sample space $\Omega = \{1, 2, 3, \dots\}$, where sample point m corresponds to 'first Head occurs on the mth flip'
- Event space $\mathcal{F}$ = all subsets of Ω
- Specify probability measure $\mathbb{P}$ by specifying the value of $\mathbb{P}(\{\omega\})$ for every $\omega \in \Omega$

In this case, Ω is a *countable infinite* set. Since any subset A of a countable set Ω is also countable, then for any $A \in \mathcal{F}$, by Property (III), we have $\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\{\omega\})$.

For example, we could set $\mathbb{P}(m) = \left(\frac{1}{2}\right)^m$ for $m = 1, 2, \dots$
(Remark: This corresponds to the *assumption* that the coin is 'fair' — see later.)

Check: $\mathbb{P}(1) + \mathbb{P}(2) + \mathbb{P}(3) + \cdots = \sum_{m=1}^{\infty} \left(\frac{1}{2}\right)^m = 1$ (using the formula for summing geometric series).

Let $A = \{2, 4, 6, \dots\} = \{\text{first Head occurs on an even flip}\}$. Then by Property (III) and the summation formula for geometric series, we have

$$\begin{aligned}
\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\{\omega\}) &= \mathbb{P}(2) + \mathbb{P}(4) + \mathbb{P}(6) + \cdots \text{ by property (III)} \\
&= \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \cdots \\
&= \frac{1}{4}\left(1 + \frac{1}{4} + \left(\frac{1}{4}\right)^2 + \dots\right) \\
&= \frac{1}{4}\left(\frac{1}{1 - \frac{1}{4}}\right) \text{ using formula for sum of geometric series} \\
&= \frac{1}{3}
\end{aligned}$$

Then can compute, for example,

$$\begin{aligned}
\mathbb{P}(\text{first Head occurs on an odd flip}) &= \mathbb{P}(A^c) \\
&= 1 - \mathbb{P}(A) \text{ by Law of Complements} \\
&= 1 - \frac{1}{3} = \frac{2}{3}
\end{aligned}$$

When the sample space is countable we can still take $\mathcal{F}$ = all subsets of Ω, and we can still define the probability measure $\mathbb{P}$ by specifying the value of $\mathbb{P}(\{\omega\})$ for every $\omega \in \Omega$ s.t. $\sum_{\omega \in \Omega} \mathbb{P}(\{\omega\}) = 1$.

When we move beyond countable sample spaces we need to be much more careful!

Example 1.5. Suppose we want to model the time spent waiting for a bus to arrive. Let ω be the time we have to wait for a bus from the time we arrive at the bus stop. What would be our sample space?

A natural choice would be $\Omega = [0, \infty)$, note this is an uncountable set! How can we construct an event space for this example?

If we attempt to construct $\mathcal{F}$ as in the countable setting we have issues. Imagine that you went to the bus stop now, there is a small probability that the next bus will arrive in 10 minutes to the nearest minute. The probability that the next bus will arrive in 10 minutes to the nearest second is extremely small, The probability that the next bus will arrive at exactly 10 minutes is zero. Indeed, for any $\omega > 0$ the probability that the bus will arrive at exactly time ω is zero. Therefore we cannot build $\mathcal{F}$ from just singleton sets. However, if we also construct the probability on intervals then it is possible, as we will see in the following example.

Example 1.6. Random experiment: Choose a number 'at random' from the interval [0, 1] in such a way that any number is 'equally likely' to be chosen. For example consider what happens when you type `runif()` in R.

Probability model:

- Sample space $\Omega = [0, 1]$ — an uncountable set.

- Event space. If we try taking $\mathcal{F}$ to be 'all subsets of Ω', then it turns out to be impossible to define a probability measure $\mathbb{P}$ on $\mathcal{F}$ that satisfies properties (I)–(III). When Ω is uncountable there are 'too many' subsets of Ω; there is no probability measure $\mathbb{P}$ on all subsets of Ω which satisfies Properties (I)–(III).

  Instead, we define the event space $\mathcal{F}$ and the probability measure $\mathbb{P}$ using some 'basic sets'.

  - Take the 'basic sets' of [0, 1] to be closed intervals of the form [a, b]. (NB - This is not the only possible choice for 'basic sets'.)
  - Let event space $\mathcal{F}$ = all subsets of [0, 1] that can be generated from the 'basic sets' using the set operations of countable unions, countable intersections, and complement.
  - The name of this event space is the Borel σ-algebra on $[0, 1]$.

- Specify probability measure $\mathbb{P}$ by specifying the value of $\mathbb{P}(B)$ for all basic sets B. Then compute $\mathbb{P}(A)$ for any $A \in \mathcal{F}$ using the properties (I)–(III) of probability measures and the definition of $\mathbb{P}$ on basic sets.

  For this example, specify probability measure $\mathbb{P}$ on the *basic sets* to reflect our assumption that any point in the interval $[0, 1]$ is *equally likely.*

  e.g.

$$\mathbb{P}\left(\left[\frac{1}{2}, \frac{3}{4}\right]\right) = \mathbb{P}\left(\text{chosen number is between } \frac{1}{2} \text{ and } \frac{3}{4}\right)$$
$$= \text{length of the interval } \left[\frac{1}{2}, \frac{3}{4}\right] = \frac{1}{4}$$

  More generally, we take $\mathbb{P}([a, b]) = b - a$ for any a, b with $0 \leqslant a \leqslant b \leqslant 1$.

Examples

1. $[0, \frac{1}{2}) \cup (\frac{3}{4}, 1] = \underbrace{[\frac{1}{2}, \frac{3}{4}]^c}_{\text{complement of a basic set}} \in \mathcal{F}$

$$\mathbb{P}\left([0, \tfrac{1}{2}) \cup (\tfrac{3}{4}, 1]\right) = \mathbb{P}\left([\tfrac{1}{2}, \tfrac{3}{4}]^c\right) = 1 - \mathbb{P}\left([\tfrac{1}{2}, \tfrac{3}{4}]\right) = 1 - \tfrac{1}{4} = \tfrac{3}{4}$$

   by Law of Complements and definition of $\mathbb{P}$ on the basic set $[\frac{1}{2}, \frac{3}{4}]$.

2. $\underbrace{\left[\frac{1}{4},\frac{1}{2}\right]\cup\left[\frac{2}{3},\frac{3}{4}\right]}_{\text{union of two basic sets}}\in\mathcal{F}$

$\mathbb{P}\left([\frac{1}{4},\frac{1}{2}]\cup[\frac{2}{3},\frac{3}{4}]\right)=\mathbb{P}\left([\frac{1}{4},\frac{1}{2}]\right)+\mathbb{P}\left([\frac{2}{3},\frac{3}{4}]\right)=\frac{1}{4}+\frac{1}{12}=\frac{1}{3}$

by Property (III) of probability measures and the definition of $\mathbb{P}$ on the basic sets $[\frac{1}{4},\frac{1}{2}]$ and $[\frac{2}{3},\frac{3}{4}]$.

3. $(\frac{1}{2},\frac{3}{4})=\left([0,\frac{1}{2}]\cup[\frac{3}{4},1]\right)^c\in\mathcal{F}$

$$\begin{aligned}\mathbb{P}\left(\left(\frac{1}{2},\frac{3}{4}\right)\right) &= 1-\mathbb{P}\left(\left[0,\frac{1}{2}\right]\cup\left[\frac{3}{4},1\right]\right)\\ &= 1-\left(\mathbb{P}\left(\left[0,\frac{1}{2}\right]\right)+\mathbb{P}\left(\left[\frac{3}{4},1\right]\right)\right)\\ &= 1-\left(\frac{1}{2}+\frac{1}{4}\right)=\frac{1}{4}\end{aligned}$$

4. $\{\frac{1}{2}\}=\underbrace{\bigcap_{m=2}^{\infty}[\frac{1}{2}-\frac{1}{m},\ \frac{1}{2}+\frac{1}{m}]}_{\text{countable intersection of basic sets}}\in\mathcal{F}$

To find $\mathbb{P}(\{\frac{1}{2}\})$, we'll use the following important result, valid for any probability measure.

Continuity of probability measure $\mathbb{P}$

Suppose $A_1\subseteq A_2\subseteq\ldots$ is a nested, increasing sequence of events in $\mathcal{F}$ and let $A=\cup_{k=1}^{\infty}A_k$. Then $\mathbb{P}(A)=\lim_{k\to\infty}\mathbb{P}(A_k)$.

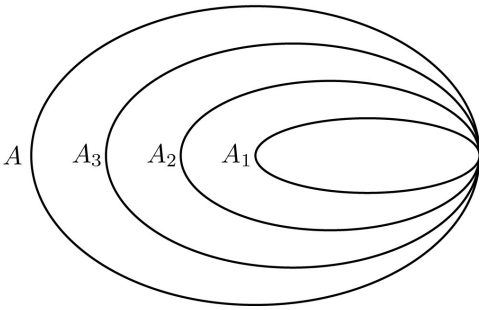


Figure 1.3: An illustration of a series of nested increasing sets A_1, A_2, A_3.

Corollary: Suppose $B_1\supseteq B_2\supseteq\ldots$ is a nested, decreasing sequence of events in $\mathcal{F}$ and let $B=\cap_{k=1}^{\infty}B_k$. Then $\mathbb{P}(B)=\lim_{k\to\infty}\mathbb{P}(B_k)$.

Application of the continuity of $\mathbb{P}$:

Since $\{\frac{1}{2}\}=\bigcap_{m=2}^{\infty}[\frac{1}{2}-\frac{1}{m},\ \frac{1}{2}+\frac{1}{m}]$, intersection of nested decreasing sequence of events, then

$$\begin{aligned}\mathbb{P}\left(\left\{\frac{1}{2}\right\}\right) &= \lim_{m\to\infty}\mathbb{P}\left(\left[\frac{1}{2}-\frac{1}{m},\frac{1}{2}+\frac{1}{m}\right]\right) \qquad \text{(by continuity of } \mathbb{P}\text{)}\\ &= \lim_{m\to\infty}\frac{2}{m}=0.\end{aligned}$$

Remarks:

(i) In general, for *this probability model*, $\mathbb{P}((a,b)) = b-a$ for any $0 \leqslant a < b \leqslant 1$, and $\mathbb{P}(\{c\}) = 0$ for *any* $c \in [0,1]$.

(ii) Although not all subsets of Ω are 'events' here, in practice any subset you might be interested in will be an event. For instance, we've seen that open intervals, closed intervals, half-open intervals, individual point values, are all events. It is possible to construct a subset of $\Omega = [0,1]$ that is *not* an event, but it's quite hard work to do this.

Summary: Building probability models

Discrete case	Non-discrete case
• $\Omega = \{w_1, w_2, \dots\}$ is a *countable set*	• Ω is an *uncountable set*
• $\mathcal{F}$ = all subsets of Ω	• $\mathcal{F}$ is *not* all subsets of Ω; instead, $\mathcal{F}$ contains 'basic sets' and all other sets that can be formed by intersections, unions and complements of basic sets
• Define $\mathbb{P}(\{\omega\})$ for every $\omega \in \Omega$ s.t. $\sum_{\omega\in\Omega} \mathbb{P}(\{\omega\}) = 1$, and then use the properties of $\mathbb{P}$ to determine $\mathbb{P}(A)$ for any $A \in \mathcal{F}$	• Define $\mathbb{P}(B)$ (in a consistent way) for every basic set B and then use the properties of $\mathbb{P}$ to determine $\mathbb{P}(A)$ for any $A \in \mathcal{F}$

1.4 Conditional probability

In this section we introduce the concept of *conditional probability*:

Suppose that $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability model, and for some event $E \in \mathcal{F}$, we have $\mathbb{P}(E) > 0$. If we *know* that the event E has occurred but we don't know anything else about the outcome of the random experiment modelled by $(\Omega, \mathcal{F}, \mathbb{P})$. *Given* this information, we can calculate a *revised* probability for every other event $F \in \mathcal{F}$ which we call the *conditional probability* of F *given* the event E.

Conditional probability arises in many applications of statistics, for example if we can not observe directly the variable we are interested in. Consider a medical trial of n patients then what is the probability that a drug is effective given a proportion p patients recovered?

Before giving the definition of conditional probability let us consider the following example to motivate the definition.

Example 1.7. Suppose we flip a fair coin 3 times.
The sample space is $\Omega = \{$HHH, HHT, HTH, THH, TTH, THT, HTT, TTT$\}$.
Define events E, F to be

E = {exactly 2 flips are Heads}

F = {first flip is Heads}

'Fair' means $\mathbb{P}$(Heads) = $\mathbb{P}$(Tails) for each coin flip, and since the coin is fair, each outcome in Ω is equally likely. So we have

$$\mathbb{P}(E) = \mathbb{P}(\{\text{HHT, HTH, THH}\}) = \frac{3}{8}$$

$$\mathbb{P}(F) = \mathbb{P}(\{\text{HHH, HHT, HTH, HTT}\}) = \frac{1}{2}$$

Now suppose you know that the event E has occurred (i.e. exactly 2 flips were Heads). Since we know that E has occurred, we can essentially treat E as if it was the whole sample space. So now think of the sample space as being $E = \{\text{HHT, HTH, THH}\}$, and because the coin is fair, each outcome in E is equally likely. That is, the *conditional probabilities* are

$$\begin{aligned}\mathbb{P}(\{\text{HHT}\} \mid E) &= \frac{1}{3}\\ \mathbb{P}(\{\text{HTH}\} \mid E) &= \frac{1}{3}\\ \mathbb{P}(\{\text{THH}\} \mid E) &= \frac{1}{3}\end{aligned}$$

Also notice that outcomes that are not in E have conditional probability 0, given that E is known to have occurred. That is, $\mathbb{P}(\text{HHH} \mid E) = \mathbb{P}(\text{TTH} \mid E) = \mathbb{P}(\text{THT} \mid E) = \mathbb{P}(\text{HTT} \mid E) = \mathbb{P}(\text{TTT} \mid E) = 0$.

So we have

$$\begin{aligned}\mathbb{P}(F \mid E) &= \mathbb{P}(\{\text{HHH, HHT, HTH, HTT}\} \mid E)\\ &= \mathbb{P}(\{\text{HHH}\} \mid E) + \mathbb{P}(\{\text{HHT}\} \mid E) + \mathbb{P}(\{\text{HTH}\} \mid E) + \mathbb{P}(\{\text{HTT}\} \mid E)\\ &= 0 + \frac{1}{3} + \frac{1}{3} + 0 = \frac{2}{3}\end{aligned}$$

Notice that in this case, $\mathbb{P}(F \mid E) > \mathbb{P}(F)$. That is, knowledge that exactly 2 out of 3 flips are Heads increases the chance that the first flip is Heads.

In general (for a countable sample space), if we know that an event E has occurred, then to compute conditional probabilities:

(i) Outcomes $\omega \in E^c$ have conditional probability $\mathbb{P}(\{\omega\} \mid E) = 0$

(ii) Outcomes $\omega \in E$ have conditional probabilites computed by re-normalising the set of probabilities $\{\mathbb{P}(\{\omega\} : \omega \in E\}$ to add up to 1. That is, we multiply the original probabilities $\mathbb{P}(\{\omega\})$ by the normalising factor $\frac{1}{\mathbb{P}(E)}$ to obtain the conditional probabilites $\mathbb{P}(\{\omega\} \mid E) = \frac{\mathbb{P}(\{\omega\})}{\mathbb{P}(E)}$ for all $\omega \in E$ (exercise: check that these add up to 1).

(iii) For any event F, the conditional probability $\mathbb{P}(F \mid E)$ is computed by adding up the conditional probabilities for all outcomes $\omega \in F$. That is,

$$\begin{aligned}\mathbb{P}(F \mid E) &= \sum_{\omega \in F} \mathbb{P}(\{\omega\} \mid E)\\ &= \sum_{\omega \in E \cap F} \mathbb{P}(\{\omega\} \mid E) + \sum_{\omega \in E^c \cap F} \mathbb{P}(\{\omega\} \mid E)\\ &= \sum_{\omega \in E \cap F} \frac{\mathbb{P}(\{\omega\})}{\mathbb{P}(E)} + 0 \text{ using (i) and (ii) above}\\ &= \frac{\sum_{\omega \in E \cap F} \mathbb{P}(\{\omega\})}{\mathbb{P}(E)}\\ &= \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(E)}\end{aligned}$$

Formally, we *define* conditional probability as follows (for any sample space, countable or uncountable).

Definition
Suppose $E, F \in \mathcal{F}$ are two events, and $\mathbb{P}(E) > 0$. Then the *conditional probability* $\mathbb{P}(F|E)$ of

the event F given that the event E has occurred is defined as

$$\boxed{\mathbb{P}(F|E) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(E)}}$$

Example 1.7 continued. From the definition of conditional probability, we compute

$$\mathbb{P}(F|E) = \frac{\mathbb{P}(F \cap E)}{\mathbb{P}(E)} = \frac{\mathbb{P}(\{\text{HHT, HTH}\})}{\mathbb{P}(\{\text{HHT, HTH, THH}\})} = \frac{\frac{2}{8}}{\frac{3}{8}} = \frac{2}{3}$$

Note: The answer is the same as we obtained before, but the calculation is quicker.

Properties
Conditional probabilities satisfy the *same* properties as the unconditional probability measure $\mathbb{P}$. That is, for any event $E \in \mathcal{F}$ such that $\mathbb{P}(E) > 0$, we have

(I) $\forall A \in \mathcal{F},\ \mathbb{P}(A \mid E) \geqslant 0$

(II) $\mathbb{P}(\Omega \mid E) = 1$ and $\mathbb{P}(\emptyset \mid E) = 0$

(III) For any *mutually exclusive* events $A_1, A_2, \cdots \in \mathcal{F}$, we have $\mathbb{P}(\bigcup A_i \mid E) = \sum \mathbb{P}(A_i \mid E)$

Remark Here, the event E is one specific fixed event, whereas we consider the conditional probabilities $\mathbb{P}(A \mid E)$ for *all* events $A \in \mathcal{F}$. Sometimes indicate this by writing the *conditional probability measure* as $\mathbb{P}(\,\cdot \mid E)$

Exercise: Use the definition of conditional probability and the properties of the unconditional probability measure $\mathbb{P}$ to show that properties (I)–(III) above hold for the conditional probability measure $\mathbb{P}(\cdot \mid E)$, for any event E such that $\mathbb{P}(E) > 0$.

Since properties (I)-(III) above hold, all the rules of probability that can be deduced from (I)-(III) also hold. For example,

$\mathbb{P}(A \cup B \mid E) = \mathbb{P}(A \mid E) + \mathbb{P}(B \mid E) - \mathbb{P}(A \cap B \mid E)$ (Inclusion-Exclusion)
$\mathbb{P}(A \mid E) = 1 - \mathbb{P}(A^c \mid E)$ (Law of Complements)

and so on.

Rearranging the definition of conditional probability gives the following very useful formula for computing the probability of the intersection of two events.

Chain rule:
For any events $E, F \in \mathcal{F}$ with $\mathbb{P}(E) > 0$,

$$\boxed{\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F \mid E)}$$

This is sometimes referred to as the *Chain Rule* for probabilities.
Note: The Chain rule follows from the definition of conditional probability $\mathbb{P}(F|E) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(E)}$.

Example 1.8. Suppose an urn contains six balls, numbered $1, 2, 3, 4, 5, 6$, and we remove two balls, one at a time, from the urn without replacement, such that each of 30 possible outcomes are equally likely. What is the probability that the sum of the numbers on the balls is *even*?

For $i = 1, 2$, define events

O_i = Odd number on ith draw

E_i = Even number on ith draw $= O_i^c$

Then

$$\begin{aligned}\mathbb{P}(\text{sum is even}) &= \mathbb{P}(\text{both numbers are odd}) + \mathbb{P}(\text{both numbers are even}) \\ &= \mathbb{P}(O_1 \cap O_2) + \mathbb{P}(E_1 \cap E_2) \\ &= \mathbb{P}(O_1)\mathbb{P}(O_2 \mid O_1) + \mathbb{P}(E_1)\mathbb{P}(E_2 \mid E_1) \qquad \text{(Chain Rule)} \\ &= \frac{3}{6} \cdot \frac{2}{5} + \frac{3}{6} \cdot \frac{2}{5} = \frac{2}{5}\end{aligned}$$

Note: $\mathbb{P}(O_2 \mid O_1) = \frac{2}{5}$ since, given that the first ball chosen is odd and it is removed, there are then 2 odd balls left in the remaining 5 balls. Similarly, $\mathbb{P}(E_2 \mid E_1) = \frac{2}{5}$.

(We could alternatively have solved the above example by noting that there are 6×5 possible outcomes in this experiment and all are equally likely, so each has probability $\frac{1}{30}$. So we could count how many outcomes result in an even sum, and multiply by $\frac{1}{30}$. However, it's a bit fiddly to count all the ways to get an even sum; the chain rule argument is neater.)

The chain rule can be extended to give us a method for computing intersections of more than two events in terms of conditional probabilities:

Extended Chain Rule:
The chain rule can be extended to intersections of three events:

$$\mathbb{P}(E \cap F \cap G) = \mathbb{P}(E)\mathbb{P}(F \mid E)\mathbb{P}(G \mid E \cap F)$$

and more generally:

$$\mathbb{P}(E_1 \cap E_2 \cap \cdots \cap E_n) = \mathbb{P}(E_1)\mathbb{P}(E_2 \mid E_1)\mathbb{P}(E_3 \mid E_1 \cap E_2) \ldots \mathbb{P}(E_n \mid E_1 \cap \cdots \cap E_{n-1})$$

The following theorem is an important application of the Chain Rule.

Law of Total Probability (Partition Theorem)
Suppose events $E_1, E_2, \ldots$ form a *partition* of Ω — that is, events $E_1, E_2, \ldots$ are disjoint with $\Omega = \cup_i E_i$, and $\mathbb{P}(E_i) > 0 \ \forall i$. Then for *any* event $F \in \mathcal{F}$,

$$\boxed{\mathbb{P}(F) = \mathbb{P}(E_1)\mathbb{P}(F \mid E_1) + \mathbb{P}(E_2)\mathbb{P}(F \mid E_2) + \cdots = \sum_i \mathbb{P}(E_i)\mathbb{P}(F \mid E_i)}$$

Proof
Suppose $E_1, E_2, \ldots$ form a partition of Ω. Then for any $F \in \mathcal{F}$,

$$\begin{aligned}\mathbb{P}(F) &= \mathbb{P}(F \cap \Omega) \\ &= \mathbb{P}\left(F \cap (\cup_i E_i)\right) \\ &= \mathbb{P}\left(\cup_i (F \cap E_i)\right) && \text{(Distributive Property from set theory)} \\ &= \sum_i \mathbb{P}(F \cap E_i) && \left(\begin{smallmatrix}\text{property (III) of probability measures} \\ \text{since the events } F \cap E_1, F \cap E_2, \ldots \text{ are disjoint}\end{smallmatrix}\right) \\ &= \sum_i \mathbb{P}(E_i)\mathbb{P}(F \mid E_i) && \text{(chain rule)}\end{aligned}$$

Example 1.9. The probability that a randomly chosen individual will develop a particular disease is 10^{-4}. Call this event A, so $\mathbb{P}(A) = 10^{-4}$. There is a test for this disease which returns the result Y (positive) or N (negative), and it is known that $\mathbb{P}(Y \mid A) = 0.95$, and that $\mathbb{P}(Y \mid A^c) = 0.1$. An individual is selected at random from the population.

(i) What is the probability that this individual tests positive?

Since the events A and A^c form a *partition*, then by the Law of Total Probability,

$$\mathbb{P}(Y) = \mathbb{P}(Y \mid A)\mathbb{P}(A) + \mathbb{P}(Y \mid A^c)\mathbb{P}(A^c) = 0.95 \cdot 10^{-4} + 0.1 \cdot (1 - 10^{-4}) = 0.100085$$

(ii) Now suppose the individual *has* tested positive. What is the probability that the individual will develop the disease?

Using the definition of conditional probability and then the chain rule,

$$\mathbb{P}(A \mid Y) = \frac{\mathbb{P}(A \cap Y)}{\mathbb{P}(Y)} = \frac{\mathbb{P}(Y \mid A)\mathbb{P}(A)}{\mathbb{P}(Y)} = \frac{0.95 \times 0.0001}{0.100085} = 0.00095$$

This is about 10 times more likely than the unconditional probability $\mathbb{P}(A) = 0.0001$ but is still very small. A person who has tested positive is still unlikely to develop the disease.

The second calculation above illustrates another very useful formula for computing probabilities: for any two events A and B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}$$

Using the Law of Total Probability together with the definition of conditional probability, we obtain the following generalisation.

Bayes Theorem
Suppose events $E_1, E_2, \ldots$ form a partition of Ω with $\mathbb{P}(E_i) > 0 \ \forall i$. Then for *any* event $F \in \mathcal{F}$, for each i,

$$\boxed{\mathbb{P}(E_i \mid F) = \frac{\mathbb{P}(E_i)\mathbb{P}(F \mid E_i)}{\sum_i \mathbb{P}(E_i)\mathbb{P}(F \mid E_i)}}$$

Proof:

$$\begin{aligned}
\mathbb{P}(E_i \mid F) &= \frac{\mathbb{P}(E_i \cap F)}{\mathbb{P}(F)} && \text{(definition of conditional probability)} \\
&= \frac{\mathbb{P}(E_i)\mathbb{P}(F \mid E_i)}{\sum_i \mathbb{P}(E_i)\mathbb{P}(F \mid E_i)} && \text{(chain rule + Law of Total Probability)}
\end{aligned}$$

1.5 Independence

Statistical independence between events is defined as follows.

Definition
Two events E and F are independent with respect to a probability measure $\mathbb{P}$ if and only if

$$\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F)$$

Meaning of statistical independence
Suppose E, F are independent events with $\mathbb{P}(E) > 0$. Then we have

$$\mathbb{P}(F \mid E) = \frac{\mathbb{P}(F \cap E)}{\mathbb{P}(E)} \overset{\text{independence}}{=} \frac{\mathbb{P}(F)\mathbb{P}(E)}{\mathbb{P}(E)} = \mathbb{P}(F)$$

That is, if events E and F are independent, then the *conditional probability of F given E* is the *same* as the unconditional probability of F. Similarly, provided $\mathbb{P}(F) > 0$, then $\mathbb{P}(E \mid F) = \mathbb{P}(E)$. So for independent events, knowledge that one event has occurred does not affect the probability that the other has occurred.

Example 1.10. Suppose a die is rolled once, so $\Omega = \{1, 2, 3, 4, 5, 6\}$. Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6\}$. Suppose the die is *fair*. Are the events A and B independent?

Solution:
Since the die is fair, $\mathbb{P}(\{i\}) = \frac{1}{6} \quad \forall i \in \Omega$. To check whether the definition of independence is satisfied for events A and B, first notice that $A \cap B = \{2, 4\}$. Then

$$\mathbb{P}(A) = \frac{4}{6} = \frac{2}{3}$$
$$\mathbb{P}(B) = \frac{3}{6} = \frac{1}{2}$$
$$\mathbb{P}(A \cap B) = \mathbb{P}(\{2, 4\}) = \frac{2}{6} = \frac{1}{3}$$

Since $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, the events A and B are independent.

Now suppose we have a *biased* die, and the probabilities for each outcome are:

$$\mathbb{P}(1) = \mathbb{P}(2) = \mathbb{P}(3) = \mathbb{P}(4) = \mathbb{P}(5) = \frac{1}{10}, \quad \mathbb{P}(6) = \frac{1}{2}$$

Are the events A and B independent for this biased die?

Check whether the definition of independence is satisfied:

$$\mathbb{P}(A) = \mathbb{P}(1) + \mathbb{P}(2) + \mathbb{P}(3) + \mathbb{P}(4) = \frac{4}{10}$$
$$\mathbb{P}(B) = \mathbb{P}(2) + \mathbb{P}(4) + \mathbb{P}(6) = \frac{1}{10} + \frac{1}{10} + \frac{1}{2} = \frac{7}{10}$$
$$\mathbb{P}(A \cap B) = \mathbb{P}(2) + \mathbb{P}(4) = \frac{2}{10}$$

In this case, $\mathbb{P}(A \cap B) = \frac{2}{10} \neq \frac{4}{10} \cdot \frac{7}{10} = \mathbb{P}(A)\mathbb{P}(B)$, so the events A and B are not independent.

This example illustrates that we cannot determine whether events are independent without computing *probabilities*, i.e. whether two events are independent depends on the probability measure.

The definition of independence can be generalised as follows.

Definition
The events $E_1, E_2, \ldots$ are independent if and only if for every finite collection of events $E_{i_1}, \ldots, E_{i_k}$,

$$\mathbb{P}(E_{i_1} \cap E_{i_2} \cap \cdots \cap E_{i_k}) = \mathbb{P}(E_{i_1})\mathbb{P}(E_{i_2}) \ldots \mathbb{P}(E_{i_k})$$

Using independence to construct probability models
In practice, when building probability models we often make the *model assumption* (based on physical considerations) that certain outcomes (or events) are independent. We then use this independence assumption to build the model. The next example illustrates this.

Example 1.11. Suppose that we flip a *biased* coin 3 times and that each time we flip the coin, $\mathbb{P}(\text{Heads}) = \frac{1}{3}$ and $\mathbb{P}(\text{Tails}) = \frac{2}{3}$.

Since the outcome on one flip cannot physically affect the outcome on any other flip (that is, flips are *physically independent*), we will *assume* that the flips are statistically independent.

The sample space is $\Omega = \{\text{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}\}$, and we define the probability measure $\mathbb{P}$ using the assumption that the outcomes on different flips are independent, so

$$\mathbb{P}(\text{HHH}) \underset{\text{by indep of flips}}{=} \mathbb{P}(\text{H})\mathbb{P}(\text{H})\mathbb{P}(\text{H}) = \frac{1}{3}\cdot\frac{1}{3}\cdot\frac{1}{3} = \left(\frac{1}{3}\right)^3$$

$$\mathbb{P}(\text{HHT}) \underset{\text{by indep of flips}}{=} \mathbb{P}(\text{H})\mathbb{P}(\text{H})\mathbb{P}(\text{T}) = \left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)$$

and so on.

Warning
A common mistake is to apply the formula $\mathbb{P}(E\cap F) = \mathbb{P}(E)\mathbb{P}(F)$ to events E, F that are not independent — this will give the wrong answer. To avoid this, always start from the more fundamental formula $\mathbb{P}(E\cap F) = \mathbb{P}(E)\mathbb{P}(F \mid E)$, and then think carefully about whether you are justified in replacing $\mathbb{P}(F \mid E)$ with $\mathbb{P}(F)$; this is only justified if you know that events E, F are independent.